

SOFTWARE REQUIREMENT SPECIFICATION

Vani Tuition Centre System

Group 4 | May 2026

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Section 1 — Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the Vani Tuition Centre System. The system is designed to digitalise tuition centre operations by integrating student management, teacher management, attendance tracking, assignment handling, marks management, notifications, and AI learning support into a single web-based platform. This document serves as a reference for developers, stakeholders, administrators, and project supervisors during system development.

1.2 Scope

System Name: Vani Tuition Centre System

The system shall provide:

- Student Management
- Teacher Management
- QR Attendance Management
- Assignment Management
- Marks Management
- Notification Management
- AI Learning Support
- Parent Monitoring Features
- Payment Status Tracking

The system shall NOT provide:

- Online payment gateway
- Multi-branch tuition management
- Video conferencing classes
- SMS gateway integration (if not planned)

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
Admin	System administrator managing operations
Teacher	Staff member handling assignments and marks
Student	Tuition centre learner
Parent	Guardian monitoring student progress
QR	Quick Response code used for attendance

AI	Artificial Intelligence
SRS	Software Requirement Specification
API	Application Programming Interface

1.4 Document Overview

This document is organised into six sections as follows:

- Section 1 – Introduction
- Section 2 – Overall Description
- Section 3 – System Analysis
- Section 4 – Functional Requirements
- Section 5 – Non-Functional Requirements
- Section 6 – Alternative Solutions & Feasibility Study

1.5 References

Software Technologies

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AI Learning Support

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- Artificial Intelligence Concepts and Applications in Education.

Software Engineering

- Sommerville, I. Software Engineering, 10th Edition. Pearson Education, 2015.
- Pressman, R. S., and Maxim, B. R. Software Engineering: A Practitioner's Approach, 9th Edition. McGraw-Hill Education.
- IEEE Software Requirements Specification Standard.

Project Documentation

- Client Interview Records – Vani Tuition Centre Requirements Gathering
- Project Use Case Diagrams
- Activity Diagrams
- Functional Requirement Specifications
- Non-Functional Requirement Specifications

Section 2 — Overall Description

2.1 Project Scope

The Vani Tuition Centre System is a web-based platform developed to digitalise and centralise tuition centre operations. The system aims to improve administrative efficiency, communication, attendance tracking, assignment handling, and student learning support.

Included Features

- Student record management
- Teacher record management
- QR code attendance management
- Assignment upload and submission management
- Marks upload and viewing
- SMS notification service for parents
- Payment status monitoring
- Attendance report generation
- AI chatbot learning support
- Learning resource management
- Parent meeting notifications
- General announcement notifications

Excluded Features

- Online payment gateway integration
- Video conferencing facilities
- Multi-branch tuition centre management
- Biometric attendance integration
- Mobile application support (if not planned)

2.2 Stakeholders

Stakeholder	Role	Responsibility
Admin	System Administrator	Manage students, teachers, notifications, attendance reports, learning resources
Teacher	Academic Staff	Upload assignments, review submissions, upload marks
Student	System User	View assignments, submit assignments, view marks, access AI chatbot
Parent	Guardian	Receive SMS notifications, monitor student progress
QR Attendance System	External System	Validate attendance QR scans

AI Chatbot System	External System	Provide AI learning assistance
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2.3 Stakeholder Requirements

Admin Requirements

- Manage student records
- Manage teacher records
- Send SMS notifications
- View attendance reports
- Upload learning resources

Teacher Requirements

- Upload assignments
- Review student submissions
- Upload marks

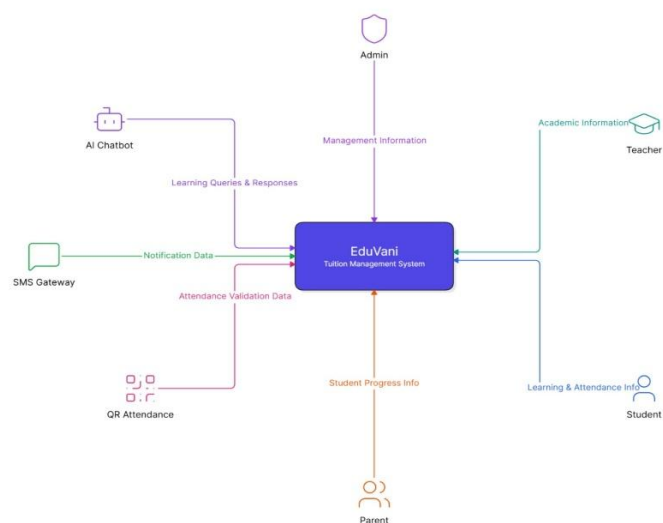
Student Requirements

- Scan attendance QR
- Submit assignments
- View assignments
- View marks
- Access AI chatbot

Parent Requirements

- Receive SMS notifications
- View student marks

2.4 Context Diagram



2.5 Personas

User personas represent the primary stakeholders of the Vani Tuition Centre System, illustrating their goals, behaviours, and pain points to guide system design decisions.

Persona 1 — Admin (System Administrator)

Name & Role: Nimal Perera — Tuition Centre Administrator

Goals and Motivations:

- Manage all student and teacher records from a single centralised platform
- Monitor attendance, payment status, and generate reports quickly
- Send notifications to parents without manual phone calls
- Reduce paperwork and administrative errors in daily operations
- Maintain full control over system access and data security

Key Pain Points with Current System:

- Spends excessive time maintaining paper-based registers for students and teachers
- No automated way to notify parents — must call or message each parent individually
- Tracking overdue payments is difficult without a digital record system
- Generating attendance or marks reports requires manual compilation from multiple sources
- Risk of data loss due to physical record keeping

Expected Interactions with EduVani:

- Log in and access a role-based admin dashboard
- Add, update, search, and delete student and teacher records
- View attendance reports and payment statuses at a glance
- Send SMS notifications (attendance, payment reminders, meeting notices, announcements) in a few clicks
- Upload learning resources for the AI chatbot

- Monitor all system activity from a centralised interface
-

Persona 2 — Teacher

Name & Role: Priya Jayasinghe — Subject Teacher (Mathematics)

Goals and Motivations:

- Upload assignments to students digitally without distributing physical papers
- Review student submissions efficiently within the system
- Upload marks after assessments and make them accessible to students and parents
- Focus more on teaching rather than administrative coordination

Key Pain Points with Current System:

- Assignments must be physically distributed and collected, causing coordination delays
- No structured way to track which students have submitted and which have not
- Marks are recorded manually in registers, making it difficult to share with parents quickly
- Communication between teacher, admin, and parents relies entirely on manual effort

Expected Interactions with EduVani:

- Log in and access a teacher dashboard showing assigned subjects and pending tasks
 - Upload assignments with title, subject, description, and deadline
 - View student submissions and provide feedback or review through the system
 - Enter and upload student marks by class or assessment
 - Access only the modules and data relevant to assigned subjects
-

Persona 3 — Student

Name & Role: Kasun Fernando — O/L Student (Age 15)

Goals and Motivations:

- Access assignments and submit them digitally without needing to be physically present

- View marks after assessments without waiting for manual distribution
- Get help on subject questions at any time using AI learning support
- Track attendance and stay updated on tuition centre announcements

Key Pain Points with Current System:

- Assignments are given and collected only during class sessions — missing a class means missing the assignment
- No way to check marks without waiting for the teacher to manually share them
- No learning support available outside of class hours
- Attendance records are not visible to students or parents in real time

Expected Interactions with EduVani:

- Log in and view all published assignments from the student dashboard
- Upload and submit assignment files before the deadline
- View marks for each subject and assessment after the teacher uploads them
- Scan QR code to record attendance during class sessions
- Access the AI chatbot to ask subject-related questions and receive answers based on uploaded learning resources

Persona 4 — Parent

Name & Role: Sandya Fernando — Parent (Mother of Kasun, O/L Student)

Goals and Motivations:

- Stay informed about her child's attendance in real time
- Receive timely notifications about payments, meetings, and announcements
- Monitor her child's academic performance without needing to visit the tuition centre
- Feel confident that her child is attending regularly and performing well

Key Pain Points with Current System:

- Only finds out about attendance issues when the child mentions it — no proactive notification

- Payment reminders are not communicated clearly or on time
- Has no access to marks unless the child physically brings the mark sheet home
- Parent meeting notices are communicated informally and sometimes missed

Expected Interactions with EduVani:

- Receive an SMS notification automatically when her child scans attendance
- Receive SMS reminders for upcoming or overdue payments
- Receive SMS notices for parent meetings and general announcements
- Log in to view her child's marks for each subject and assessment
- Have a clear, up-to-date view of her child's academic and attendance status at any time

2.6 Operating Environment

Environment Factor	Description
Web Browser	Chrome, Edge, Firefox, Safari
Operating System	Windows and Linux environments
Backend Runtime	Spring Boot (Java)
Frontend Framework	React
Database	MySQL
Network	Internet connection required (HTTPS)
Server	Cloud or local server deployment
Mobile	Responsive design for desktop, tablet, and mobile devices

2.7 Business Rules

Student Management Rules

ID	Business Rule
BR-001	Each student shall have a unique student identifier
BR-002	Only Admin can manage student records

BR-003	Student attendance records shall remain linked to student profiles
BR-004	Student payment status shall remain associated with student accounts

Teacher Management Rules

ID	Business Rule
BR-005	Each teacher shall have a unique teacher identifier
BR-006	Only Admin can manage teacher records
BR-007	Teachers can only upload marks and assignments for assigned subjects
BR-008	Teacher payment information shall remain restricted to Admin access

Attendance Rules

ID	Business Rule
BR-009	QR attendance validation must occur before attendance recording
BR-010	Attendance records cannot be manually modified without Admin permission
BR-011	Attendance notifications shall be sent to parents via SMS
BR-012	Attendance reports shall contain accurate attendance information

Notification Rules

ID	Business Rule
BR-013	All parent notifications shall be delivered through SMS notifications
BR-014	Only Admin can send general announcements
BR-015	Payment reminders shall contain accurate payment information
BR-016	Parent meeting notices shall contain meeting date, time, and location

Assignment Rules

ID	Business Rule
BR-017	Only Teachers can upload assignments
BR-018	Students shall submit assignments before deadlines
BR-019	Assignment submissions shall store submission timestamps
BR-020	Teachers can review only available student submissions

Marks Rules

ID	Business Rule
BR-021	Only Teachers can upload marks
BR-022	Students can view only their own marks

BR-023	Parents can view only marks related to their registered child
BR-024	Marks information shall remain protected from unauthorized modification

AI Learning Rules

ID	Business Rule
BR-025	AI chatbot responses shall use uploaded learning resources
BR-026	Only authenticated students can access AI learning support
BR-027	Learning resources shall remain manageable only by Admin

2.8 System Constraints

ID	Constraint
CON-001	System shall operate as a web-based application
CON-002	Internet connectivity is required
CON-003	SMS gateway integration is required for parent notifications
CON-004	QR code scanning device support is required
CON-005	System shall use MySQL database
CON-006	Backend development shall use Spring Boot
CON-007	Frontend development shall use React

2.9 Assumptions

ID	Assumption
ASM-001	Users have internet access
ASM-002	Parents possess mobile phones capable of receiving SMS notifications
ASM-003	Students possess valid QR attendance credentials
ASM-004	Teachers regularly upload assignments and marks
ASM-005	Learning resources are uploaded before AI chatbot usage
ASM-006	SMS notification service remains available during system operation

Section 3 — System Analysis

3.1 Fact-Finding Methods

3.1.1 Interview

Field	Details
Project Name	Vani Tuition Centre System
Interview Type	Structured Interview
Interview Date	23.05.2026
Interview Duration	30 – 45 Minutes
Interview Method	Face-to-Face / Online Meeting
Interview Participant	Tuition Centre Administrator (Client)
Interview Conducted By	Group 4 team members
Purpose	To gather functional and non-functional requirements for the Vani Tuition Centre System

The following structured questions were used during the interview:

Student Management

- How are student records currently managed?
- What student information should be stored?
- Who should have permission to manage student records?
- Should attendance and payment status be linked to student profiles?

Teacher Management

- What teacher information should the system maintain?
- How should teacher subject assignments be managed?
- Should teacher payment records be maintained?

Attendance Management

- How should attendance be recorded?
- Should QR code attendance be used?
- Who should receive attendance notifications?

Notification System

- What types of notifications should parents receive?
- Should overdue payment reminders be automated?
- Should general announcements and parent meeting notices be supported?

Assignment Management

- Should teachers upload assignments digitally?
- Should students submit assignments through the system?
- Should teachers review submissions within the system?

Marks Management

- How should marks be uploaded?
- Who should access student marks?
- Should parents be allowed to view marks?

AI Learning Support

- Should AI chatbot support be included?
- What type of learning resources should be uploaded?
- Should students ask subject-related questions using AI?

Interview Summary

The client explained that current tuition centre operations are mostly managed manually, causing difficulty in maintaining student information, attendance tracking, payment monitoring, assignment handling, and communication with parents. The client requested a centralised web-based solution capable of managing student records, teacher records, QR-based attendance tracking, notification delivery, assignment handling, marks management, and AI-powered learning assistance. The client emphasised the importance of parent communication and attendance monitoring, and highlighted the need for AI chatbot support to improve student learning experiences.

Client Pain Points

Pain Point	Description
Manual record keeping	Student and teacher records maintained using paper-based methods, causing data loss and inefficiency
Attendance tracking	Attendance marked manually without digital record, making reporting difficult
Payment monitoring	No system to track or remind parents of overdue payments
Parent communication	No automated notification mechanism for parent communication
Assignment handling	Assignments distributed and submitted physically without digital tracking
Marks management	Marks recorded in manual registers without secure digital storage or easy parent access
Learning support	No AI or digital learning support available for students

Interview Key Findings – Functional Requirements

ID	Requirement
FR-001	Student record management
FR-002	Teacher record management

FR-003	QR attendance management
FR-004	Parent notification system
FR-005	Assignment upload and submission
FR-006	Marks upload and viewing
FR-007	AI chatbot learning support
FR-008	Payment status monitoring
FR-009	Teacher payment management
FR-010	Attendance report generation

Interview Key Findings – Non-Functional Requirements

ID	Requirement
NFR-001	User authentication
NFR-002	Secure data storage
NFR-003	Fast system response
NFR-004	User-friendly interface
NFR-005	Reliable notification delivery
NFR-006	System availability
NFR-007	Database consistency

3.1.2 Observation

Purpose

The observation technique was used to study the actual working environment of the tuition centre and validate the information collected through interviews.

Method

Direct observation was carried out during regular tuition centre operations. Activities such as student registration, attendance marking, assignment handling, marks entry, and parent communication were carefully observed.

Observations

- Attendance was recorded manually during each class session.
- Student and teacher information was stored in physical registers and spreadsheets.
- Assignment distribution and collection required significant manual coordination.
- Marks were recorded manually before being shared with students and parents.
- Parent notifications were sent individually, leading to delays and inefficiency.

Findings

- Manual processes increased the risk of data entry errors.
- Administrative workload was high due to repetitive tasks.

- Reporting and information retrieval were time-consuming.
- Lack of centralised data storage reduced efficiency and consistency.

Outcome

The observation confirmed the issues identified in the interview and provided additional insights into real-time operational inefficiencies. These findings strongly justify the need for the EduVani system.

3.2 Existing System Analysis

The following section presents a detailed analysis of the existing manual system currently in use at the Vani Tuition Centre, based on findings from the client interview and direct observation.

Current Workflows

The tuition centre currently operates entirely through manual processes. Student and teacher records are maintained using physical registers and paper files stored at the centre. Attendance is marked manually by the teacher or admin during each class session using a paper-based class register. Assignments are printed or handwritten and distributed physically during class, with submissions collected in person by the teacher. Marks are recorded by hand in mark registers after assessments and shared with students verbally or through written mark sheets. Parent communication is carried out through individual phone calls or manually written messages. Payment tracking is managed using handwritten ledgers maintained by the admin.

Current Data Flows

- Student registration information is written into a physical student register by the admin.
 - Attendance data is recorded in a paper class register during each session and stored physically.
 - Assignment details are communicated verbally or on paper; no submission tracking exists.
 - Marks data flows from the teacher's mark register to the student verbally or via paper — parents receive this information indirectly through the student.
 - Payment information is written into a ledger by admin; no automated flow exists to notify parents of dues.
 - No data flows digitally between any of these processes — each operates in isolation.
-

Limitations and Inefficiencies

Area	Limitation
Record Management	Physical registers are prone to damage, loss, and human error
Attendance Tracking	No digital backup; retrieval and reporting is time-consuming
Assignment Handling	No tracking of who received, submitted, or missed assignments
Marks Management	No secure storage; parents depend on students to relay marks information
Parent Communication	Manual and individual; causes significant delays and missed notifications
Payment Tracking	No automated reminders; overdue payments are easy to overlook
Data Centralisation	No central database; information is scattered across multiple physical sources
Report Generation	Requires manual compilation from multiple registers; slow and error-prone
Scalability	System cannot scale as student numbers grow without increasing manual workload

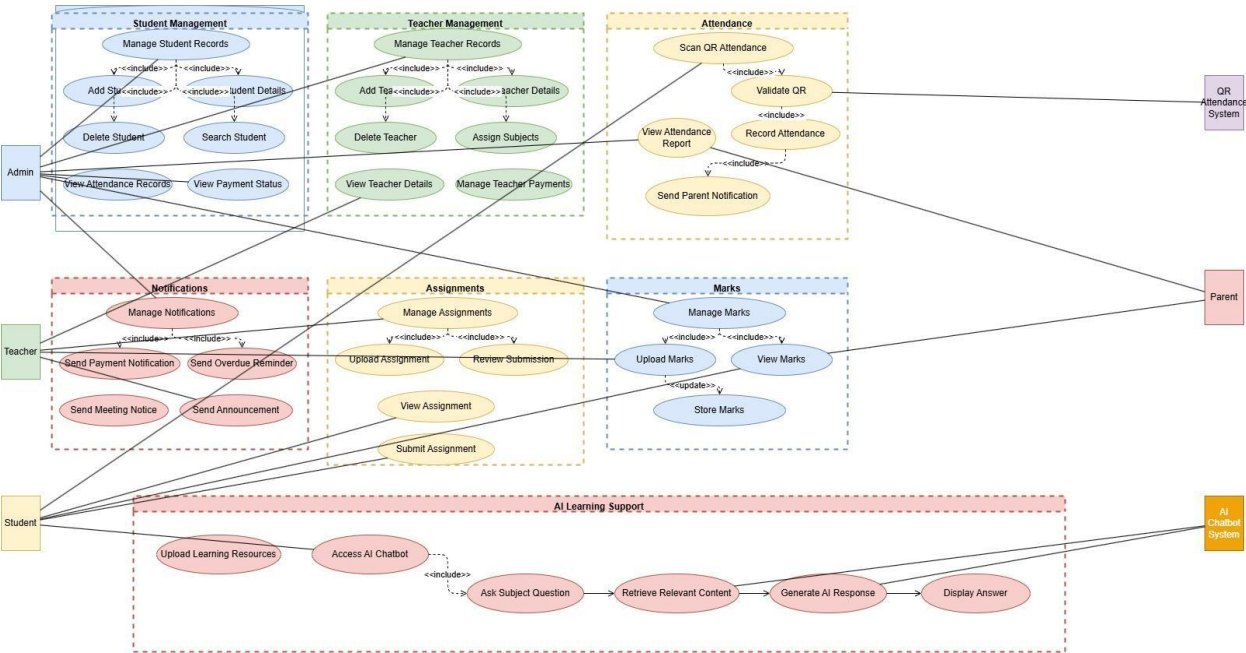
Note on Process Maps and Data Flow Diagrams

Process maps and data flow diagrams (DFDs) of the existing system should be drawn and inserted here to visually represent the current workflows described above. Recommended diagrams include:

- A context-level DFD showing information flows between Admin, Teacher, Student, and Parent in the current manual system
- A level-1 DFD breaking down each functional area (registration, attendance, assignments, marks, payments, notifications)
- A process map showing the end-to-end student registration and attendance workflow as it currently exists

3.3 Use Case Diagrams

The following use case diagram represents the complete use case model of the Vani Tuition Centre System, showing all actors including Admin, Teacher, Student, Parent, QR Attendance System, and AI Chatbot System.



[Figure 3.1: Vani Tuition Centre System – Use Case Diagram]

3.4 Use Case Descriptions

UC-01 Manage Student Records

Use Case ID	UC-01
Use Case Name	Manage Student Records
Actor(s)	Admin
Preconditions	Admin is authenticated and logged into the system.
Postconditions – Success	Student information is successfully maintained and database is updated.
Postconditions – Failure	Student operation is not completed and existing records remain unchanged.
Main Flow	1. Admin accesses the Student Management module. 2. System displays student management options. 3. Admin selects required operation. 4. System requests required information. 5. Admin enters information. 6. System validates entered information. 7. System updates or retrieves database records. 8. System displays operation result confirmation.
Alternative Flow 1	At Step 6: If entered information is invalid, system displays validation message and requests correction.

Alternative Flow 2	At Step 7: If database access fails, system displays operation failure notification.
Business Rules	BR-01: Student ID must be unique. BR-02: Mandatory fields cannot remain empty. BR-03: Only Admin can manage student records.

UC-02 Add Student

Use Case ID	UC-02
Use Case Name	Add Student
Actor(s)	Admin
Preconditions	Admin is authenticated. Student does not already exist in the system.
Postconditions – Success	New student record is successfully added to the database.
Postconditions – Failure	Student record is not created and database remains unchanged.
Main Flow	1. Admin selects Add Student option. 2. System displays student registration form. 3. Admin enters student information. 4. System validates entered information. 5. System stores student information in database. 6. System displays successful registration confirmation.
Alternative Flow 1	At Step 4: If required information is missing, system displays validation error message.
Alternative Flow 2	At Step 5: If student ID already exists, system displays duplicate record notification.
Business Rules	BR-01: Student ID must be unique. BR-02: Required fields cannot remain empty. BR-03: Student contact information must follow valid format.

UC-03 Update Student Details

Use Case ID	UC-03
Use Case Name	Update Student Details
Actor(s)	Admin
Preconditions	Admin is authenticated. Student record already exists in the system.
Postconditions – Success	Student information is successfully updated and database reflects latest changes.
Postconditions – Failure	Student information is not updated and existing data remains unchanged.
Main Flow	1. Admin selects Update Student Details. 2. System requests student identification. 3. Admin searches and selects the student record. 4. System displays existing student details. 5. Admin modifies required information. 6. System validates updated information. 7. System updates database records. 8. System displays update confirmation.

Alternative Flow 1	At Step 3: If student record cannot be found, system displays student not found message.
Alternative Flow 2	At Step 6: If entered information is invalid, system displays validation error and requests correction.
Business Rules	BR-01: Student ID cannot be modified. BR-02: Mandatory fields cannot remain empty. BR-03: Only Admin can update student information.

UC-04 Delete Student

Use Case ID	UC-04
Use Case Name	Delete Student
Actor(s)	Admin
Preconditions	Admin is authenticated. Student record exists in the database.
Postconditions – Success	Student record is successfully removed from the system database.
Postconditions – Failure	Student record is not deleted and database remains unchanged.
Main Flow	1. Admin selects Delete Student. 2. System requests student identification. 3. Admin searches and selects student record. 4. System displays selected student information. 5. Admin confirms deletion request. 6. System removes student record from database. 7. System displays successful deletion confirmation.
Alternative Flow 1	At Step 3: If student record cannot be found, system displays student not found message.
Alternative Flow 2	At Step 5: If Admin cancels, system terminates operation without removing data.
Business Rules	BR-01: Only Admin can delete student records. BR-02: Deleted records cannot be recovered unless backup is available. BR-03: System must request confirmation before removing records.

UC-05 Search Student

Use Case ID	UC-05
Use Case Name	Search Student
Actor(s)	Admin
Preconditions	Admin is authenticated. Student records are available in the database.
Postconditions – Success	Requested student information is successfully displayed.
Postconditions – Failure	Requested student information is not found and no records are displayed.

Main Flow	1. Admin selects Search Student. 2. System displays search interface. 3. Admin enters search criteria. 4. System validates criteria. 5. System searches database. 6. System retrieves matching information. 7. System displays search results.
Alternative Flow 1	At Step 4: If search criteria is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If no matching record is found, system displays student not found notification.
Business Rules	BR-01: Only Admin can search student records. BR-02: Search must use valid criteria only. BR-03: System should display accurate and updated information.

UC-06 View Attendance Records

Use Case ID	UC-06
Use Case Name	View Attendance Records
Actor(s)	Admin
Preconditions	Admin is authenticated. Attendance records are available in the database.
Postconditions – Success	Attendance records are successfully retrieved and displayed.
Postconditions – Failure	Attendance records cannot be retrieved and system displays an error.
Main Flow	1. Admin selects View Attendance Records. 2. System displays attendance search interface. 3. Admin enters search criteria. 4. System validates information. 5. System retrieves attendance records. 6. System displays attendance information. 7. Admin reviews attendance details.
Alternative Flow 1	At Step 4: If entered information is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If attendance records are unavailable, system displays record not found notification.
Business Rules	BR-01: Only Admin can access attendance records. BR-02: Information must display updated records. BR-03: Records must be associated with valid student information.

UC-07 View Payment Status

Use Case ID	UC-07
Use Case Name	View Payment Status
Actor(s)	Admin
Preconditions	Admin is authenticated. Payment records are available in the database.
Postconditions – Success	Payment information is successfully retrieved and displayed.

Postconditions – Failure	Payment information cannot be retrieved and system displays an error.
Main Flow	1. Admin selects View Payment Status. 2. System displays payment search interface. 3. Admin enters search criteria. 4. System validates information. 5. System retrieves payment records. 6. System displays payment information. 7. Admin reviews payment details.
Alternative Flow 1	At Step 4: If entered information is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If payment records are unavailable, system displays payment record not found notification.
Business Rules	BR-01: Only Admin can access payment information. BR-02: Payment status must display updated records. BR-03: Payment records must be associated with valid student information.

UC-08 Manage Teacher Records

Use Case ID	UC-08
Use Case Name	Manage Teacher Records
Actor(s)	Admin
Preconditions	Admin is authenticated and logged into the system.
Postconditions – Success	Teacher information is successfully maintained and database is updated.
Postconditions – Failure	Teacher record operation is not completed and existing records remain unchanged.
Main Flow	1. Admin accesses Teacher Management module. 2. System displays teacher management options. 3. Admin selects required operation. 4. System requests required information. 5. Admin enters information. 6. System validates entered information. 7. System updates or retrieves teacher records. 8. System displays operation result confirmation.
Alternative Flow 1	At Step 6: If entered information is invalid, system displays validation error and requests correction.
Alternative Flow 2	At Step 7: If database access fails, system displays operation failure notification.
Business Rules	BR-01: Only Admin can manage teacher records. BR-02: Teacher information must contain required fields. BR-03: System must maintain updated teacher information.

UC-09 Add Teacher

Use Case ID	UC-09
Use Case Name	Add Teacher
Actor(s)	Admin

Preconditions	Admin is authenticated. Teacher record does not already exist.
Postconditions – Success	New teacher record is successfully added to the database.
Postconditions – Failure	Teacher record is not created and database remains unchanged.
Main Flow	1. Admin selects Add Teacher. 2. System displays teacher registration form. 3. Admin enters teacher information. 4. System validates information. 5. System stores teacher information. 6. System displays successful registration confirmation.
Alternative Flow 1	At Step 4: If required information is missing or invalid, system displays validation error.
Alternative Flow 2	At Step 5: If teacher record already exists, system displays duplicate record notification.
Business Rules	BR-01: Teacher ID must be unique. BR-02: Required fields cannot remain empty. BR-03: Only Admin can add teacher records.

UC-10 Update Teacher Details

Use Case ID	UC-10
Use Case Name	Update Teacher Details
Actor(s)	Admin
Preconditions	Admin is authenticated. Teacher record already exists.
Postconditions – Success	Teacher information is successfully updated and database reflects latest changes.
Postconditions – Failure	Teacher information is not updated and existing records remain unchanged.
Main Flow	1. Admin selects Update Teacher Details. 2. System requests teacher identification. 3. Admin searches and selects teacher record. 4. System displays existing teacher details. 5. Admin modifies required information. 6. System validates updated information. 7. System updates teacher records. 8. System displays update confirmation.
Alternative Flow 1	At Step 3: If teacher record cannot be found, system displays teacher not found message.
Alternative Flow 2	At Step 6: If entered information is invalid, system displays validation error and requests correction.
Business Rules	BR-01: Teacher ID cannot be modified. BR-02: Required fields cannot remain empty. BR-03: Only Admin can update teacher information.

UC-11 Delete Teacher

Use Case ID	UC-11
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Use Case Name	Delete Teacher
Actor(s)	Admin
Preconditions	Admin is authenticated. Teacher record exists in the database.
Postconditions – Success	Teacher record is successfully removed from the system database.
Postconditions – Failure	Teacher record is not deleted and database remains unchanged.
Main Flow	1. Admin selects Delete Teacher. 2. System requests teacher identification. 3. Admin searches and selects teacher record. 4. System displays selected teacher information. 5. Admin confirms deletion. 6. System removes teacher record. 7. System displays successful deletion confirmation.
Alternative Flow 1	At Step 3: If teacher record cannot be found, system displays teacher not found message.
Alternative Flow 2	At Step 5: If Admin cancels, system terminates operation without removing data.
Business Rules	BR-01: Only Admin can delete teacher records. BR-02: System must request confirmation before removing records. BR-03: Teacher record must exist before deletion can occur.

UC-12 Assign Subjects

Use Case ID	UC-12
Use Case Name	Assign Subjects
Actor(s)	Admin
Preconditions	Admin is authenticated. Teacher record exists. Subject information is available.
Postconditions – Success	Selected subjects are successfully assigned to the teacher and database is updated.
Postconditions – Failure	Subject assignment is not completed and existing information remains unchanged.
Main Flow	1. Admin selects Assign Subjects. 2. System displays available teachers. 3. Admin selects a teacher record. 4. System displays available subject list. 5. Admin selects required subjects. 6. System validates selected information. 7. System assigns subjects to teacher. 8. System displays assignment confirmation.
Alternative Flow 1	At Step 3: If selected teacher record is unavailable, system displays teacher not found message.
Alternative Flow 2	At Step 6: If selected subject information is invalid, system displays validation error.
Business Rules	BR-01: Only Admin can assign subjects. BR-02: Teacher record must exist before assigning subjects. BR-03: Subject assignments must be updated in the database immediately.

UC-13 View Teacher Details

Use Case ID	UC-13
Use Case Name	View Teacher Details
Actor(s)	Admin
Preconditions	Admin is authenticated. Teacher records are available in the database.
Postconditions – Success	Teacher information is successfully retrieved and displayed.
Postconditions – Failure	Teacher information cannot be retrieved and system displays an error.
Main Flow	1. Admin selects View Teacher Details. 2. System displays teacher search interface. 3. Admin enters teacher identification or search criteria. 4. System validates information. 5. System retrieves teacher records. 6. System displays teacher information. 7. Admin reviews teacher details.
Alternative Flow 1	At Step 4: If entered information is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If teacher record is unavailable, system displays teacher not found notification.
Business Rules	BR-01: Only Admin can access teacher information. BR-02: Teacher information displayed must be updated and accurate. BR-03: Teacher records must exist before viewing details.

UC-14 Manage Teacher Payments

Use Case ID	UC-14
Use Case Name	Manage Teacher Payments
Actor(s)	Admin
Preconditions	Admin is authenticated. Teacher payment information is available in the database.
Postconditions – Success	Teacher payment information is successfully updated and stored in the database.
Postconditions – Failure	Teacher payment operation is not completed and existing records remain unchanged.
Main Flow	1. Admin selects Manage Teacher Payments. 2. System displays payment management interface. 3. Admin searches and selects teacher record. 4. System displays teacher payment information. 5. Admin enters or updates payment details. 6. System validates payment information. 7. System updates payment records. 8. System displays payment update confirmation.
Alternative Flow 1	At Step 3: If teacher record cannot be found, system displays teacher not found message.

Alternative Flow 2	At Step 6: If payment information is invalid, system displays validation error and requests correction.
Business Rules	BR-01: Only Admin can manage teacher payments. BR-02: Payment information must contain valid values. BR-03: Teacher payment records must be stored and updated accurately.

UC-15 Scan QR Attendance

Use Case ID	UC-15
Use Case Name	Scan QR Attendance
Actor(s)	Student
Preconditions	Student is authenticated. QR attendance code is available and active.
Postconditions – Success	Student attendance scanning request is successfully submitted for validation.
Postconditions – Failure	Attendance scanning request fails and attendance is not recorded.
Main Flow	1. Student accesses attendance feature. 2. System displays QR scanning interface. 3. Student scans QR attendance code. 4. System captures QR information. 5. System sends QR information for validation. 6. System proceeds to attendance validation process.
Alternative Flow 1	At Step 3: If QR code cannot be scanned, system displays scanning error message.
Alternative Flow 2	At Step 4: If QR information is incomplete, system requests rescan.
Business Rules	BR-01: Student must use a valid QR code. BR-02: Attendance QR code must be active during class session. BR-03: Student attendance scan must occur only once per session.

UC-16 Validate QR

Use Case ID	UC-16
Use Case Name	Validate QR
Actor(s)	QR Attendance System
Preconditions	Student has scanned the QR code. QR attendance information has been received.
Postconditions – Success	QR code is successfully validated and attendance recording process continues.
Postconditions – Failure	QR validation fails and attendance is not recorded.
Main Flow	1. System receives scanned QR information. 2. System sends QR data to QR Attendance System. 3. QR Attendance System checks QR validity. 4. QR Attendance System verifies session information. 5. QR Attendance

	System returns validation result. 6. System proceeds to attendance recording process.
Alternative Flow 1	At Step 3: If QR code is invalid, system displays invalid QR notification.
Alternative Flow 2	At Step 4: If QR session has expired, system displays expired QR notification and terminates attendance process.
Business Rules	BR-01: Only active QR codes can be validated. BR-02: QR validation must occur before recording attendance. BR-03: Expired QR codes cannot be accepted.

UC-17 Record Attendance

Use Case ID	UC-17
Use Case Name	Record Attendance
Actor(s)	QR Attendance System
Preconditions	QR code validation is successfully completed. Student attendance information is available.
Postconditions – Success	Student attendance is successfully recorded in the database.
Postconditions – Failure	Attendance record is not stored and system displays an error.
Main Flow	1. System receives successful QR validation result. 2. System retrieves student information. 3. System prepares attendance record. 4. System stores attendance information in database. 5. System updates attendance status. 6. System proceeds to parent notification process.
Alternative Flow 1	At Step 2: If student information cannot be retrieved, system displays student record error.
Alternative Flow 2	At Step 4: If database storage fails, system displays attendance recording failure notification.
Business Rules	BR-01: Attendance can only be recorded after successful QR validation. BR-02: Attendance records must contain valid student information. BR-03: System must store attendance records immediately after validation.

UC-18 Send Parent Notification

Use Case ID	UC-18
Use Case Name	Send Parent Notification
Actor(s)	Parent
Preconditions	Student attendance has been successfully recorded. Parent contact information is available.
Postconditions – Success	Attendance notification is successfully delivered to the parent.

Postconditions – Failure	Notification delivery fails and system records notification failure status.
Main Flow	1. System detects successful attendance recording. 2. System retrieves parent contact information. 3. System generates attendance notification message. 4. System sends notification to parent. 5. Parent receives attendance notification. 6. System records notification delivery status.
Alternative Flow 1	At Step 2: If parent contact information is unavailable, system displays parent information error.
Alternative Flow 2	At Step 4: If notification delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Parent notification can only be sent after attendance recording. BR-02: Notification must contain accurate attendance information. BR-03: Notification must be sent only to registered parent contact details.

UC-19 View Attendance Report

Use Case ID	UC-19
Use Case Name	View Attendance Report
Actor(s)	Admin
Preconditions	Admin is authenticated. Attendance records are available in the database.
Postconditions – Success	Attendance report is successfully generated and displayed.
Postconditions – Failure	Attendance report cannot be generated and system displays an error.
Main Flow	1. Admin selects View Attendance Report. 2. System displays attendance report interface. 3. Admin enters required report criteria. 4. System validates entered criteria. 5. System retrieves attendance records. 6. System generates attendance report. 7. System displays attendance report. 8. Admin reviews attendance information.
Alternative Flow 1	At Step 4: If entered criteria is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If attendance data is unavailable, system displays attendance record unavailable notification.
Business Rules	BR-01: Only Admin can view attendance reports. BR-02: Reports must contain updated attendance information. BR-03: Report generation must use valid attendance records only.

UC-20 Manage Notifications

Use Case ID	UC-20
Use Case Name	Manage Notifications
Actor(s)	Admin

Preconditions	Admin is authenticated. Notification module is available.
Postconditions – Success	Selected notification is successfully created and delivered to intended recipients.
Postconditions – Failure	Notification operation fails and intended recipients do not receive notification.
Main Flow	1. Admin accesses Notification Management module. 2. System displays available notification options. 3. Admin selects notification type. 4. System requests required notification information. 5. Admin enters notification details. 6. System validates entered information. 7. System processes notification request. 8. System sends notification to recipients. 9. System displays notification delivery confirmation.
Alternative Flow 1	At Step 6: If entered notification information is invalid, system displays validation error.
Alternative Flow 2	At Step 8: If notification delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Only Admin can manage notifications. BR-02: Notification information must contain required details. BR-03: Notifications must only be delivered to registered recipients.

UC-21 Send Payment Notification

Use Case ID	UC-21
Use Case Name	Send Payment Notification
Actor(s)	Parent
Preconditions	Admin is authenticated. Payment information is available. Parent contact information is registered.
Postconditions – Success	Payment notification is successfully delivered to the parent.
Postconditions – Failure	Payment notification delivery fails and notification status is recorded as unsuccessful.
Main Flow	1. Admin selects Send Payment Notification. 2. System retrieves student payment information. 3. System generates payment notification message. 4. System retrieves parent contact information. 5. System sends payment notification. 6. Parent receives payment notification. 7. System records notification delivery status.
Alternative Flow 1	At Step 2: If payment information is unavailable, system displays payment information error.
Alternative Flow 2	At Step 5: If notification delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Payment notifications must contain accurate payment details. BR-02: Notifications can only be sent to registered parent contact information. BR-03: System must record notification delivery status.

UC-22 Send Overdue Payment Reminder

Use Case ID	UC-22
Use Case Name	Send Overdue Payment Reminder
Actor(s)	Parent
Preconditions	Admin is authenticated. Student payment due information is available. Parent contact information is registered.
Postconditions – Success	Overdue payment reminder is successfully delivered to the parent.
Postconditions – Failure	Reminder notification delivery fails and notification status is recorded as unsuccessful.
Main Flow	1. Admin selects Send Overdue Payment Reminder. 2. System identifies overdue payment records. 3. System retrieves parent contact information. 4. System generates overdue payment reminder message. 5. System sends reminder notification. 6. Parent receives overdue payment reminder. 7. System records notification delivery status.
Alternative Flow 1	At Step 2: If overdue payment records are unavailable, system displays no overdue payment information message.
Alternative Flow 2	At Step 5: If reminder delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Overdue reminders must contain accurate payment information. BR-02: Reminder notifications can only be sent to registered parent contact information. BR-03: System must maintain notification delivery records.

UC-23 Send Parent Meeting Notice

Use Case ID	UC-23
Use Case Name	Send Parent Meeting Notice
Actor(s)	Parent
Preconditions	Admin is authenticated. Parent meeting information is available. Parent contact information is registered.
Postconditions – Success	Parent meeting notice is successfully delivered to parents.
Postconditions – Failure	Meeting notice delivery fails and notification status is recorded as unsuccessful.
Main Flow	1. Admin selects Send Parent Meeting Notice. 2. System requests meeting information. 3. Admin enters meeting details (Date / Time / Location / Description). 4. System validates entered information. 5. System retrieves registered parent contact information. 6. System generates parent meeting notice. 7. System sends meeting notice notification. 8. Parent receives parent meeting notice. 9. System records notification delivery status.
Alternative Flow 1	At Step 4: If entered meeting information is invalid, system displays validation error.

Alternative Flow 2	At Step 7: If notification delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Meeting notice must contain accurate meeting information. BR-02: Notices can only be sent to registered parent contact information. BR-03: System must record notification delivery status.

UC-24 Send General Announcement

Use Case ID	UC-24
Use Case Name	Send General Announcement
Actor(s)	Parent
Preconditions	Admin is authenticated. Announcement information is available. Parent contact information is registered.
Postconditions – Success	General announcement is successfully delivered to intended recipients.
Postconditions – Failure	Announcement delivery fails and notification status is recorded as unsuccessful.
Main Flow	1. Admin selects Send General Announcement. 2. System requests announcement information. 3. Admin enters announcement details. 4. System validates entered information. 5. System retrieves registered recipient information. 6. System generates announcement notification. 7. System sends announcement notification. 8. Parent receives announcement notification. 9. System records notification delivery status.
Alternative Flow 1	At Step 4: If entered announcement information is invalid, system displays validation error.
Alternative Flow 2	At Step 7: If notification delivery fails, system records failure and displays delivery error.
Business Rules	BR-01: Announcement information must contain required details. BR-02: Notifications can only be sent to registered recipients. BR-03: System must maintain notification delivery records.

UC-25 Manage Assignments

Use Case ID	UC-25
Use Case Name	Manage Assignments
Actor(s)	Teacher
Preconditions	Teacher is authenticated. Assignment module is available.
Postconditions – Success	Assignment operation is successfully completed and assignment information is updated in the system.
Postconditions – Failure	Assignment operation fails and existing assignment information remains unchanged.

Main Flow	1. Teacher accesses Assignment Management module. 2. System displays assignment management options. 3. Teacher selects required operation. 4. System requests required information. 5. Teacher enters assignment information. 6. System validates entered information. 7. System processes assignment operation. 8. System updates assignment records. 9. System displays operation confirmation.
Alternative Flow 1	At Step 6: If entered information is invalid, system displays validation error.
Alternative Flow 2	At Step 7: If assignment operation fails, system displays operation failure notification.
Business Rules	BR-01: Only Teacher can manage assignments. BR-02: Assignment information must contain required details. BR-03: Assignment records must be stored and updated accurately.

UC-26 Upload Assignment

Use Case ID	UC-26
Use Case Name	Upload Assignment
Actor(s)	Teacher
Preconditions	Teacher is authenticated. Assignment information is available for upload.
Postconditions – Success	Assignment is successfully uploaded and stored in the system.
Postconditions – Failure	Assignment upload fails and assignment information is not stored.
Main Flow	1. Teacher selects Upload Assignment. 2. System displays assignment upload interface. 3. Teacher enters assignment information (Title / Subject / Description / Deadline). 4. Teacher uploads assignment file if required. 5. System validates entered information. 6. System stores assignment information and files. 7. System publishes assignment for students. 8. System displays successful upload confirmation.
Alternative Flow 1	At Step 5: If entered assignment information is invalid, system displays validation error.
Alternative Flow 2	At Step 6: If assignment file upload fails, system displays upload failure notification.
Business Rules	BR-01: Only Teacher can upload assignments. BR-02: Assignment must contain required information before publishing. BR-03: Uploaded assignment files must follow supported file formats.

UC-27 View Assignment

Use Case ID	UC-27
Use Case Name	View Assignment
Actor(s)	Student

Preconditions	Student is authenticated. Assignment information is available in the system.
Postconditions – Success	Assignment information is successfully retrieved and displayed to the student.
Postconditions – Failure	Assignment information cannot be retrieved and system displays an error.
Main Flow	1. Student accesses Assignment module. 2. System displays available assignments. 3. Student selects an assignment. 4. System retrieves assignment information. 5. System displays assignment details and attached files. 6. Student reviews assignment information.
Alternative Flow 1	At Step 2: If assignments are unavailable, system displays no assignment available notification.
Alternative Flow 2	At Step 4: If assignment retrieval fails, system displays assignment loading error.
Business Rules	BR-01: Only authenticated students can access assignments. BR-02: Students can only view published assignments. BR-03: Assignment information displayed must be updated and accurate.

UC-28 Submit Assignment

Use Case ID	UC-28
Use Case Name	Submit Assignment
Actor(s)	Student
Preconditions	Student is authenticated. Assignment is available and submission deadline has not passed.
Postconditions – Success	Assignment submission is successfully stored in the system.
Postconditions – Failure	Assignment submission fails and submission information is not stored.
Main Flow	1. Student accesses Assignment module. 2. System displays available assignments. 3. Student selects assignment for submission. 4. System displays submission interface. 5. Student uploads assignment submission file. 6. System validates uploaded submission. 7. System stores assignment submission. 8. System displays successful submission confirmation.
Alternative Flow 1	At Step 6: If uploaded file format is invalid, system displays validation error.
Alternative Flow 2	At Step 7: If submission deadline has expired, system rejects submission and displays deadline exceeded notification.
Business Rules	BR-01: Students can submit assignments only before deadline. BR-02: Assignment submission files must follow supported file formats. BR-03: System must record submission date and time.

UC-29 Review Submission

Use Case ID	UC-29
Use Case Name	Review Submission
Actor(s)	Teacher
Preconditions	Teacher is authenticated. Student assignment submissions are available in the system.
Postconditions – Success	Assignment submission is successfully reviewed and review information is stored.
Postconditions – Failure	Submission review process fails and review information is not updated.
Main Flow	1. Teacher accesses Assignment Management module. 2. System displays submitted assignments. 3. Teacher selects a student submission. 4. System retrieves submission information. 5. System displays submitted files and submission details. 6. Teacher reviews assignment submission. 7. Teacher enters feedback or evaluation if required. 8. System stores review information. 9. System displays successful review confirmation.
Alternative Flow 1	At Step 3: If submission is unavailable, system displays submission not found notification.
Alternative Flow 2	At Step 8: If review information cannot be stored, system displays review saving failure notification.
Business Rules	BR-01: Only Teacher can review assignment submissions. BR-02: Review information must be stored accurately. BR-03: Teachers can only review available student submissions.

UC-30 Manage Marks

Use Case ID	UC-30
Use Case Name	Manage Marks
Actor(s)	Teacher
Preconditions	Teacher is authenticated. Marks Management module is available.
Postconditions – Success	Marks operation is successfully completed and mark information is updated in the system.
Postconditions – Failure	Marks operation fails and existing mark information remains unchanged.
Main Flow	1. Teacher accesses Marks Management module. 2. System displays marks management options. 3. Teacher selects required operation. 4. System requests required mark information. 5. Teacher enters mark information. 6. System validates entered information. 7. System processes marks operation. 8. System updates mark records. 9. System displays operation confirmation.
Alternative Flow 1	At Step 6: If entered information is invalid, system displays validation error.
Alternative Flow 2	At Step 7: If mark processing fails, system displays operation failure notification.

Business Rules	BR-01: Only Teacher can manage marks. BR-02: Marks information must contain valid values. BR-03: Mark records must be stored and updated accurately.
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UC-31 Upload Marks

Use Case ID	UC-31
Use Case Name	Upload Marks
Actor(s)	Teacher
Preconditions	Teacher is authenticated. Student records and assessment information are available.
Postconditions – Success	Student marks are successfully uploaded and stored in the system.
Postconditions – Failure	Marks upload operation fails and mark information is not stored.
Main Flow	1. Teacher accesses Marks Management module. 2. System displays mark upload interface. 3. Teacher selects class or assessment. 4. System displays student list. 5. Teacher enters marks for students. 6. System validates entered marks. 7. System stores marks information. 8. System displays successful upload confirmation.
Alternative Flow 1	At Step 6: If entered marks are invalid, system displays validation error.
Alternative Flow 2	At Step 7: If mark storage fails, system displays upload failure notification.
Business Rules	BR-01: Only Teacher can upload marks. BR-02: Marks must contain valid numerical values. BR-03: Student marks must be stored accurately in the system.

UC-32 Store Marks

Use Case ID	UC-32
Use Case Name	Store Marks
Actor(s)	System
Preconditions	Teacher has successfully uploaded marks. Mark information has passed validation checks.
Postconditions – Success	Student marks are successfully stored in the database.
Postconditions – Failure	Marks are not stored and existing records remain unchanged.
Main Flow	1. System receives uploaded marks information. 2. System validates mark data. 3. System checks student record availability. 4. System prepares marks information for storage. 5. System stores marks in database. 6. System updates marks records. 7. System displays successful storage confirmation.

Alternative Flow 1	At Step 2: If marks information is invalid, system displays validation error.
Alternative Flow 2	At Step 5: If database storage fails, system displays mark storage failure notification.
Business Rules	BR-01: Marks must pass validation before storage. BR-02: Marks must be associated with valid student records. BR-03: System must store marks accurately in the database.

UC-33 View Marks

Use Case ID	UC-33
Use Case Name	View Marks
Actor(s)	Student, Parent
Preconditions	Student marks are available in the system. Student or Parent is authenticated.
Postconditions – Success	Marks information is successfully retrieved and displayed.
Postconditions – Failure	Marks information cannot be retrieved and system displays an error.
Main Flow	1. Student or Parent accesses Marks module. 2. System displays marks viewing interface. 3. Student or Parent selects required assessment or subject. 4. System validates access permissions. 5. System retrieves marks information. 6. System displays marks information. 7. Student or Parent reviews marks details.
Alternative Flow 1	At Step 4: If access permission is invalid, system displays unauthorised access notification.
Alternative Flow 2	At Step 5: If marks information is unavailable, system displays marks not found notification.
Business Rules	BR-01: Students can only view their own marks. BR-02: Parents can only view marks related to their registered child. BR-03: Marks information displayed must be accurate and updated.

UC-34 Upload Learning Resources

Use Case ID	UC-34
Use Case Name	Upload Learning Resources
Actor(s)	Admin
Preconditions	Admin is authenticated. Learning resource information is available for upload.
Postconditions – Success	Learning resources are successfully uploaded and stored in the system.

Postconditions – Failure	Learning resource upload fails and information is not stored.
Main Flow	1. Admin accesses AI Learning Support module. 2. System displays learning resource upload interface. 3. Admin enters learning resource information. 4. Admin uploads learning resource files. 5. System validates uploaded information and files. 6. System stores learning resources. 7. System updates AI learning resource database. 8. System displays successful upload confirmation.
Alternative Flow 1	At Step 5: If uploaded information is invalid, system displays validation error.
Alternative Flow 2	At Step 6: If file upload fails, system displays upload failure notification.
Business Rules	BR-01: Only Admin can upload learning resources. BR-02: Uploaded resources must follow supported file formats. BR-03: Learning resources must be stored accurately for AI retrieval.

UC-35 Access AI Chatbot

Use Case ID	UC-35
Use Case Name	Access AI Chatbot
Actor(s)	Student
Preconditions	Student is authenticated. AI Learning Support module is available.
Postconditions – Success	Student successfully accesses AI Chatbot and chatbot interface is displayed.
Postconditions – Failure	AI Chatbot access fails and system displays an error.
Main Flow	1. Student accesses AI Learning Support module. 2. System displays AI Chatbot access option. 3. Student selects AI Chatbot. 4. System verifies student access permissions. 5. System initialises AI Chatbot session. 6. System displays AI Chatbot interface. 7. Student accesses AI learning assistance features.
Alternative Flow 1	At Step 4: If student authentication fails, system displays access denied notification.
Alternative Flow 2	At Step 5: If AI Chatbot service is unavailable, system displays service unavailable message.
Business Rules	BR-01: Only authenticated students can access AI Chatbot. BR-02: AI Chatbot access requires active system connection. BR-03: System must create a valid chatbot session before interaction begins.

UC-36 Ask Subject Question

Use Case ID	UC-36
Use Case Name	Ask Subject Question

Actor(s)	Student
Preconditions	Student is authenticated. AI Chatbot session is active.
Postconditions – Success	Student question is successfully submitted to the AI system and response generation begins.
Postconditions – Failure	Question submission fails and system displays an error.
Main Flow	1. Student accesses AI Chatbot interface. 2. System displays question input area. 3. Student enters subject-related question. 4. Student submits question. 5. System validates entered question. 6. System forwards question to AI processing module. 7. System begins content retrieval process.
Alternative Flow 1	At Step 3: If question input is empty, system displays validation message requesting question input.
Alternative Flow 2	At Step 5: If question processing fails, system displays processing error notification.
Business Rules	BR-01: Only authenticated students can ask questions. BR-02: Question input cannot remain empty. BR-03: Questions must be processed through the AI learning support module.

UC-37 Retrieve Relevant Content

Use Case ID	UC-37
Use Case Name	Retrieve Relevant Content
Actor(s)	AI Chatbot System
Preconditions	Student question is successfully submitted. Learning resources are available in the system database.
Postconditions – Success	Relevant learning content is successfully retrieved for AI response generation.
Postconditions – Failure	Relevant content cannot be retrieved and response generation process cannot continue.
Main Flow	1. System receives student question from AI Chatbot module. 2. System analyses question content. 3. System identifies related subject area. 4. System searches stored learning resources. 5. System retrieves relevant content. 6. System prepares retrieved information for AI processing. 7. System forwards content to response generation module.
Alternative Flow 1	At Step 4: If related learning resources are unavailable, system displays content unavailable notification.
Alternative Flow 2	At Step 5: If content retrieval fails, system displays retrieval error message.
Business Rules	BR-01: Retrieved content must match the student's question context. BR-02: Learning resources must be retrieved only from authorised system data. BR-03: Retrieved content must be prepared before AI response generation.

UC-38 Generate AI Response

Use Case ID	UC-38
Use Case Name	Generate AI Response
Actor(s)	AI Chatbot System
Preconditions	Relevant learning content has been successfully retrieved. Student question information is available.
Postconditions – Success	AI response is successfully generated based on retrieved learning content.
Postconditions – Failure	AI response generation fails and system displays an error.
Main Flow	1. System receives retrieved learning content. 2. System analyses student question and retrieved information. 3. System processes relevant content. 4. System generates AI response. 5. System verifies generated response. 6. System prepares response for display. 7. System forwards generated response to answer display module.
Alternative Flow 1	At Step 3: If retrieved content is insufficient, system displays content limitation notification.
Alternative Flow 2	At Step 4: If AI response generation fails, system displays response generation error notification.
Business Rules	BR-01: AI response must be generated only using authorised learning resources. BR-02: Generated response must match the student's question context. BR-03: AI response must be prepared before displaying to the student.

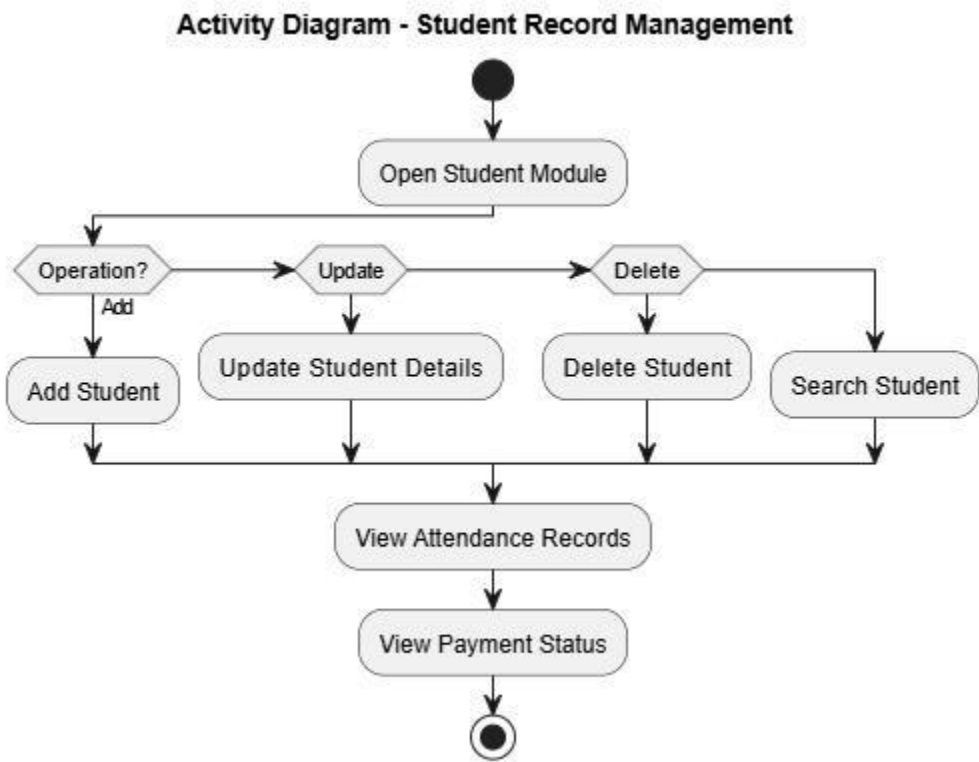
UC-39 Display Answer

Use Case ID	UC-39
Use Case Name	Display Answer
Actor(s)	Student
Preconditions	AI response has been successfully generated. Student AI Chatbot session is active.
Postconditions – Success	AI-generated answer is successfully displayed to the student.
Postconditions – Failure	Answer cannot be displayed and system shows an error notification.
Main Flow	1. System receives generated AI response. 2. System prepares answer for presentation. 3. System formats answer content for display. 4. System displays answer through AI Chatbot interface. 5. Student views generated answer. 6. Student reviews answer information. 7. System maintains chatbot session for further interaction.
Alternative Flow 1	At Step 2: If answer information is incomplete, system displays answer preparation error.

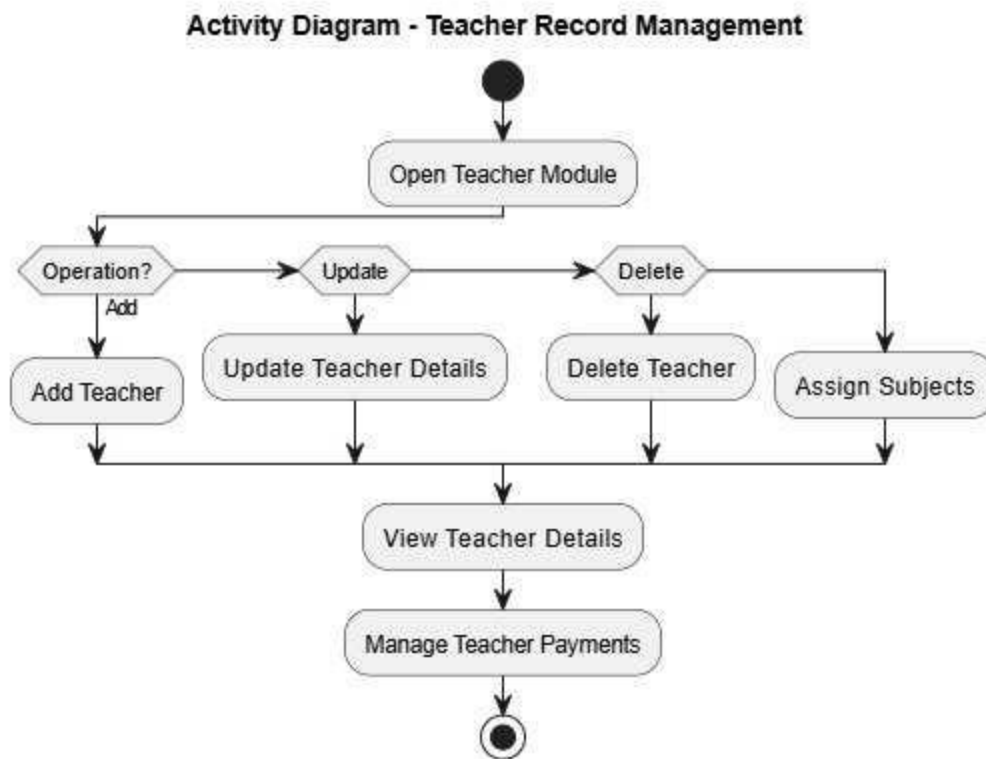
Alternative Flow 2	At Step 4: If answer display fails, system displays display error notification.
Business Rules	BR-01: Only generated AI responses can be displayed. BR-02: Displayed answer must match the student's submitted question. BR-03: Answer information must be presented clearly through the AI Chatbot interface.

3.5 Activity Diagrams

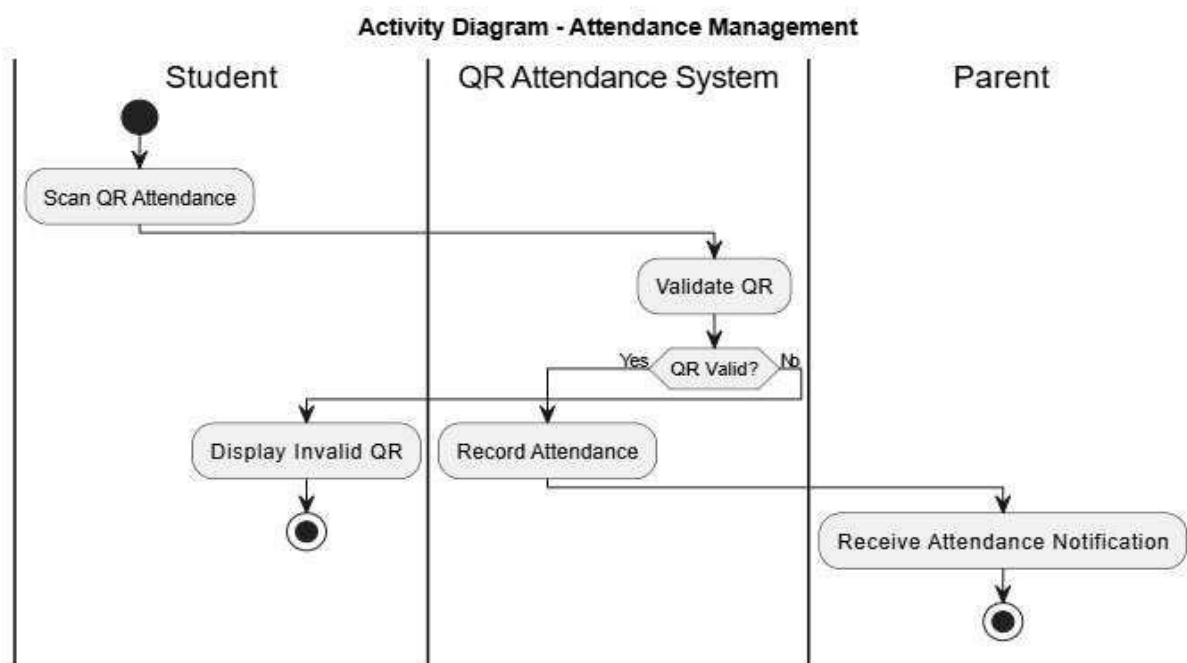
The following activity diagrams illustrate the workflow for each major functional module of the Vani Tuition Centre System.



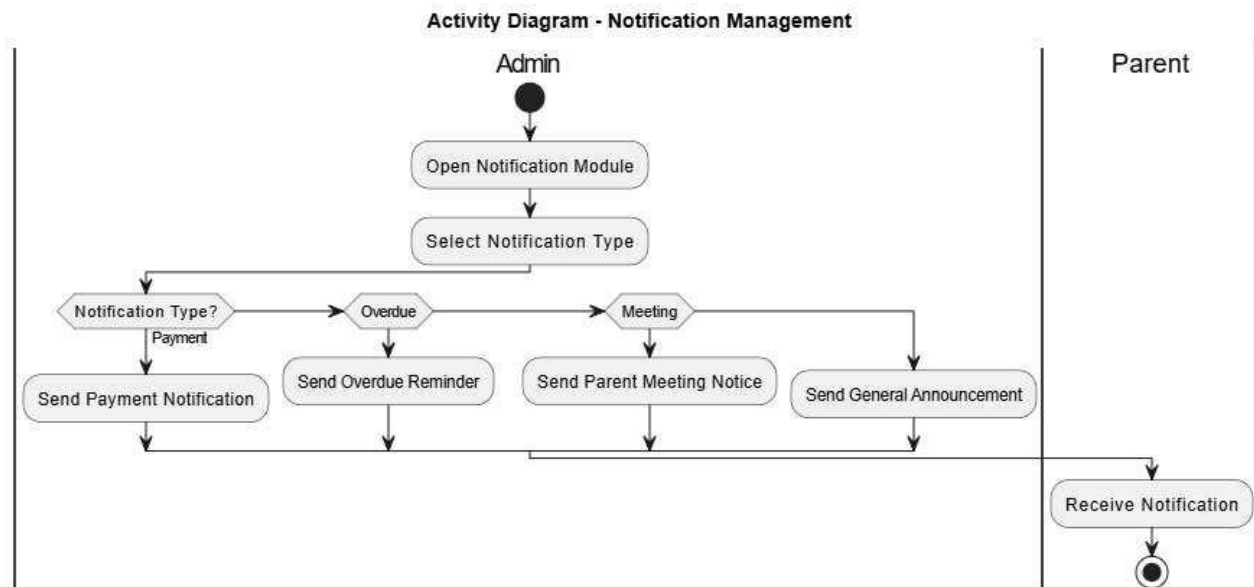
[Figure 3.2: Activity Diagram – Student Record Management]



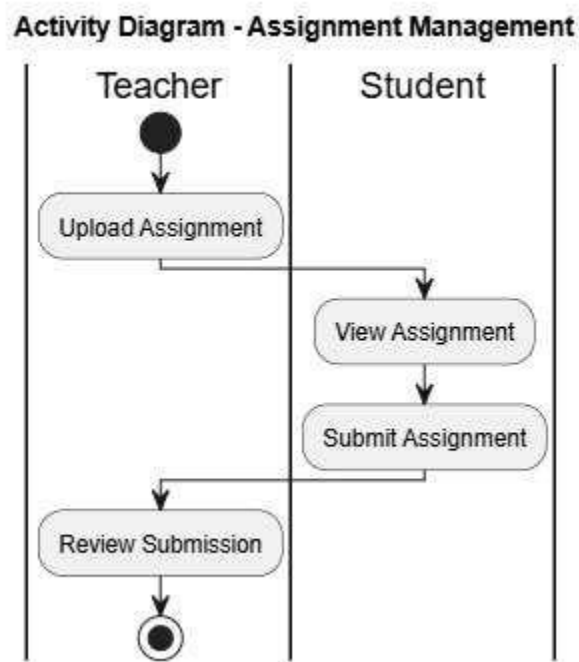
[Figure 3.3: Activity Diagram – Teacher Record Management]



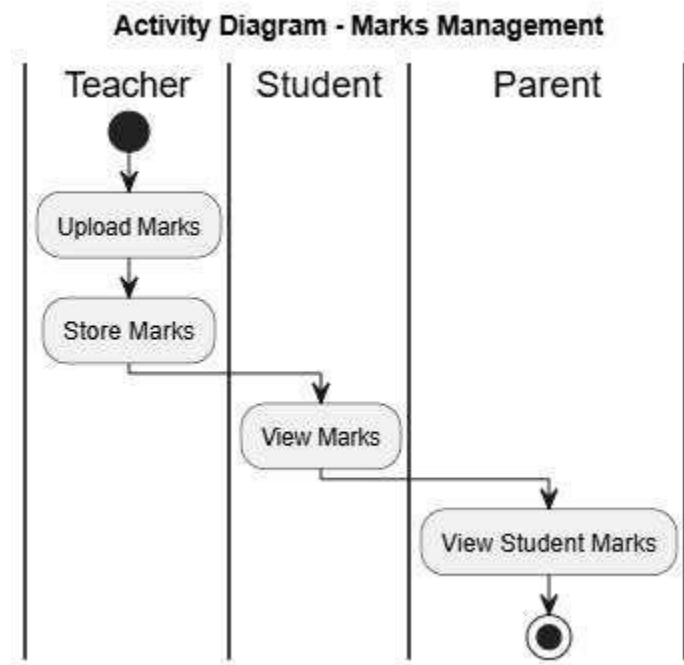
[Figure 3.4: Activity Diagram – Attendance Management]



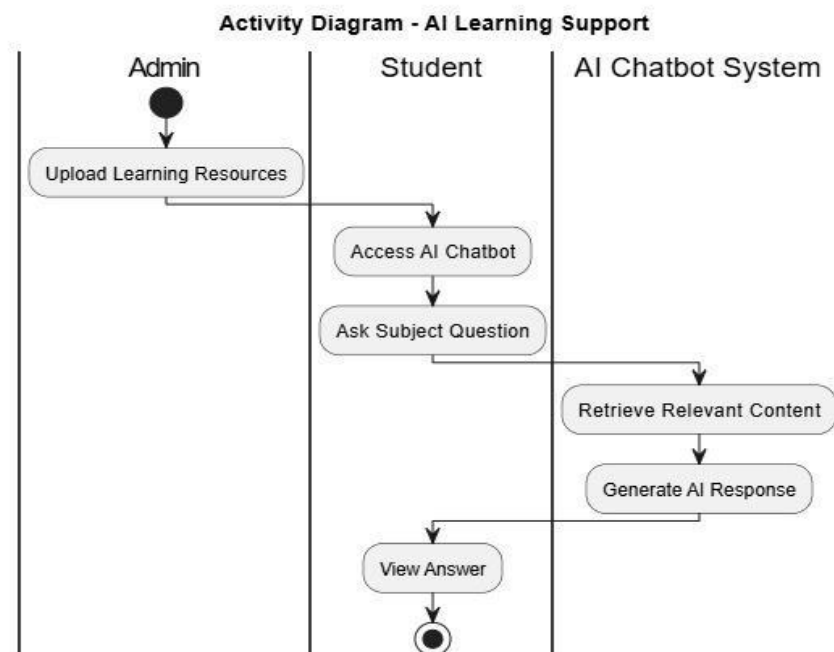
[Figure 3.5: Activity Diagram – Notification Management]



[Figure 3.6: Activity Diagram – Assignment Management]

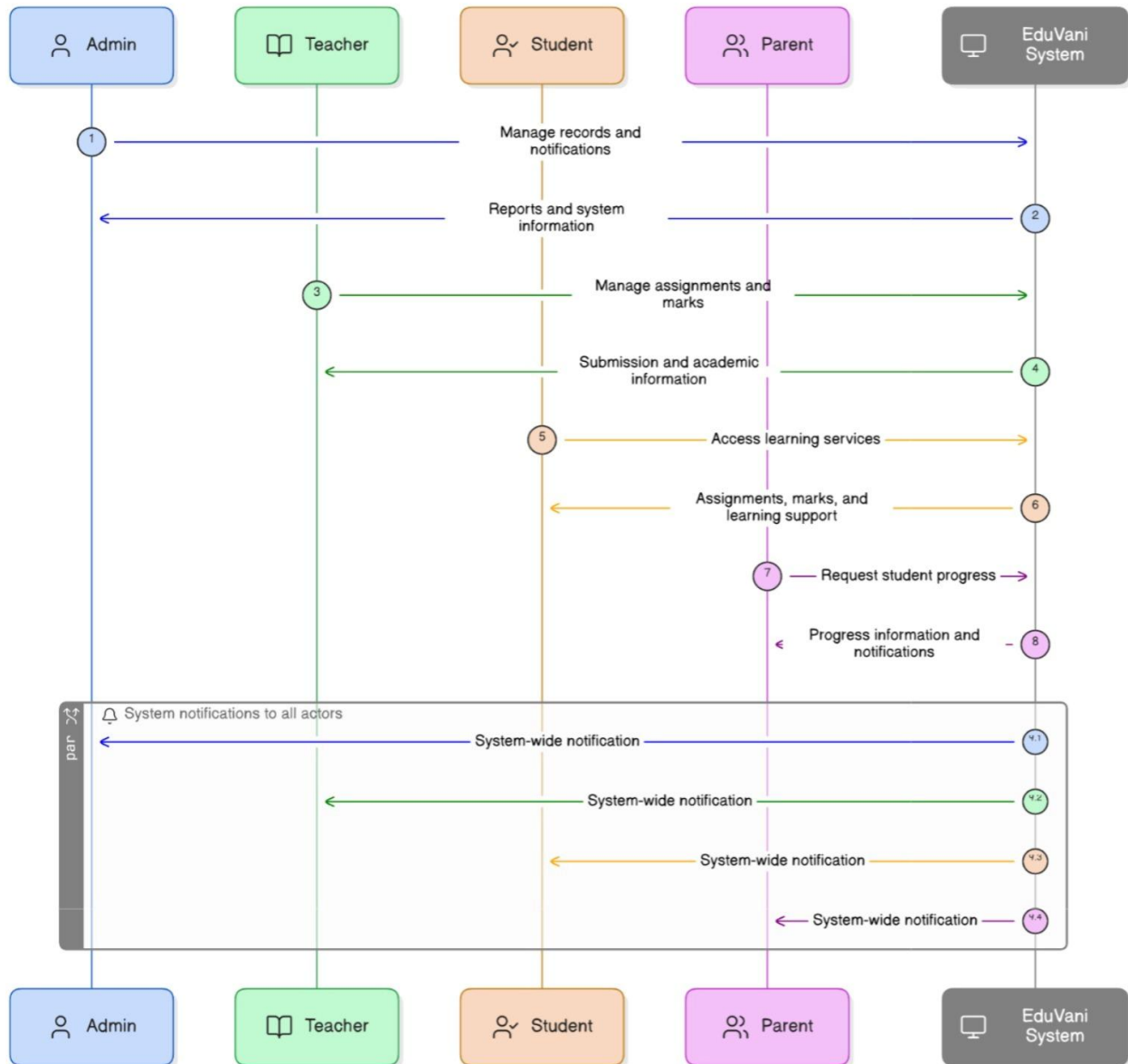


[Figure 3.7: Activity Diagram – Marks Management]



[Figure 3.8: Activity Diagram – AI Learning Support]

3.6 Sequence Diagram



Section 4 — Functional Requirements

Functional requirements define the features and operations that the Vani Tuition Centre System shall perform. Each requirement includes its Source (where the requirement originated) and Verification Method (how it will be confirmed during testing).

4.1 Student Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-001	System shall allow Admin to add student records	High	Client Interview (Student Management)	Functional testing – verify Admin can successfully add a student record
FR-002	System shall allow Admin to update student details	High	Client Interview (Student Management)	Functional testing – verify Admin can update existing student details
FR-003	System shall allow Admin to delete student records	Medium	Client Interview (Student Management)	Functional testing – verify Admin can delete a student record with confirmation
FR-004	System shall allow Admin to search student information	High	Client Interview (Student Management)	Functional testing – verify Admin can search and retrieve student records
FR-005	System shall allow Admin to view attendance records	High	Client Interview (Attendance Management)	Functional testing – verify Admin can view student attendance records
FR-006	System shall allow Admin to view student payment status	High	Client Interview (Student Management)	Functional testing – verify Admin can view accurate payment status information

4.2 Teacher Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-007	System shall allow Admin to add teacher records	High	Client Interview (Teacher Management)	Functional testing – verify Admin can successfully add a teacher record
FR-008	System shall allow Admin to update teacher information	High	Client Interview (Teacher Management)	Functional testing – verify Admin can update existing teacher details

FR-009	System shall allow Admin to delete teacher records	Medium	Client Interview (Teacher Management)	Functional testing – verify Admin can delete a teacher record with confirmation
FR-010	System shall allow Admin to assign subjects to teachers	High	Client Interview (Teacher Management)	Functional testing – verify Admin can assign subjects to teachers
FR-011	System shall allow Admin to view teacher information	High	Client Interview (Teacher Management)	Functional testing – verify Admin can retrieve and view teacher details
FR-012	System shall allow Admin to manage teacher payments	High	Client Interview (Teacher Management)	Functional testing – verify Admin can update and view teacher payment records

4.3 Attendance Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-013	System shall allow students to scan QR attendance	High	Client Interview (Attendance Management)	Functional testing – verify students can scan QR code and submit attendance
FR-014	System shall validate QR attendance information	High	Business Rules (BR-009)	Functional testing – verify QR validation rejects invalid or expired codes
FR-015	System shall record attendance information	High	Business Rules (BR-009)	Functional testing – verify attendance is stored accurately after QR validation
FR-016	System shall send attendance SMS notifications to parents	High	Client Interview (Attendance Management)	Integration testing – verify SMS notification is delivered to parent after attendance recording
FR-017	System shall allow Admin to view attendance reports	High	Client Interview (Attendance Management)	Functional testing – verify Admin can generate and view attendance reports

4.4 Notification Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-018	System shall send payment notifications via SMS	High	Client Interview (Notification System)	Integration testing – verify SMS payment

				notification is delivered to parent
FR-019	System shall send overdue payment reminders via SMS	High	Client Interview (Notification System)	Integration testing – verify overdue payment reminder is delivered to parent
FR-020	System shall send parent meeting notices via SMS	Medium	Client Interview (Notification System)	Integration testing – verify meeting notice is delivered with correct meeting details
FR-021	System shall send general announcements via SMS	Medium	Client Interview (Notification System)	Integration testing – verify general announcement is delivered to all registered parents

4.5 Assignment Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-022	Teachers shall upload assignments	High	Client Interview (Assignment Management)	Functional testing – verify Teacher can upload assignment with title, subject, description, and deadline
FR-023	Students shall view assignments	High	Client Interview (Assignment Management)	Functional testing – verify Student can view all published assignments
FR-024	Students shall submit assignments	High	Client Interview (Assignment Management)	Functional testing – verify Student can submit assignment file before deadline
FR-025	Teachers shall review assignment submissions	High	Client Interview (Assignment Management)	Functional testing – verify Teacher can access and review student submissions

4.6 Marks Management Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-026	Teachers shall upload marks	High	Client Interview (Marks Management)	Functional testing – verify Teacher can upload marks for students
FR-027	System shall store marks information	High	Business Rules (BR-021 to BR-024)	Functional testing – verify marks are stored

				accurately in the database
FR-028	Students shall view marks	High	Client Interview (Marks Management)	Functional testing – verify Student can view only their own marks
FR-029	Parents shall view student marks	High	Client Interview (Marks Management)	Functional testing – verify Parent can view marks related to their registered child only

4.7 AI Learning Support Requirements

ID	Functional Requirement	Priority	Source	Verification Method
FR-030	Admin shall upload learning resources	Medium	Business Rules (BR-025 to BR-027)	Functional testing – verify Admin can upload learning resource files
FR-031	Students shall access AI chatbot	High	Client Interview (AI Learning Support)	Functional testing – verify authenticated Student can access AI Chatbot interface
FR-032	Students shall ask subject-related questions	High	Client Interview (AI Learning Support)	Functional testing – verify Student can submit questions through AI Chatbot
FR-033	System shall retrieve relevant learning resources	High	Business Rules (BR-025)	Integration testing – verify system retrieves relevant content matching student question
FR-034	System shall generate AI responses	High	Client Interview (AI Learning Support)	Integration testing – verify AI generates a response within 5 seconds (NFR-P-005)
FR-035	System shall display AI generated answers	High	Client Interview (AI Learning Support)	Functional testing – verify generated AI answer is displayed clearly to the student

Section 5 — Non-Functional Requirements

Non-functional requirements define quality attributes and operational characteristics of the Vani Tuition Centre System. Each requirement includes its Source and Verification Method.

5.1 Performance Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-P-001	Dashboard Performance	Dashboard pages shall load within ≤ 3 seconds	High	Client Interview (General)	Performance testing – measure dashboard load time under normal load conditions
NFR-P-002	QR Attendance Response	QR attendance processing shall complete within ≤ 2 seconds	High	Client Interview (Attendance)	Performance testing – measure QR processing time from scan to confirmation
NFR-P-003	Assignment Upload Performance	Assignment upload shall complete within ≤ 5 seconds	High	Client Interview (Assignment)	Performance testing – measure upload completion time for typical assignment file sizes
NFR-P-004	SMS Notification Delivery	SMS notifications shall be delivered within ≤ 15 seconds	High	Client Interview (Notification)	Integration testing – measure SMS delivery time from trigger event to delivery
NFR-P-005	AI Response Generation	AI chatbot responses shall generate within ≤ 5 seconds	Medium	Client Interview (AI Support)	Performance testing – measure response generation time from question submission
NFR-P-006	Report Generation	Attendance and marks reports shall generate within ≤ 10 seconds	Medium	Client Interview (Reporting)	Performance testing – measure report generation time

					under normal data loads
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5.2 Reliability Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-R-001	Availability	System uptime shall be $\geq 99\%$ monthly	High	Business Operational Needs	Monitoring – track system uptime over a one-month period using server logs
NFR-R-002	Backup	System backup shall occur every 24 hours	High	Data Protection Needs	Verification – confirm backup logs show daily backup completion
NFR-R-003	Recovery	System recovery after failure shall complete within ≤ 4 hours	High	Operational Continuity Needs	Recovery testing – simulate failure and measure time to restore system
NFR-R-004	Data Consistency	Attendance, marks, and payment records shall remain consistent	High	Data Integrity Requirements	Database testing – verify records remain consistent across concurrent operations

5.3 Usability Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-U-001	Easy Learning	New users shall learn core functions within 30 minutes	High	Client Interview (Usability)	User acceptance testing – observe new users completing core tasks within 30 minutes
NFR-U-002	Mobile Responsiveness	System shall function correctly on desktop, tablet, and mobile devices	High	System Constraints (CON-001)	Compatibility testing – verify system renders correctly on

					desktop, tablet, and mobile
NFR-U-003	Navigation Efficiency	Common operations shall complete within three navigation steps	Medium	Usability Standards	Usability testing – verify common operations are reachable within three clicks
NFR-U-004	Form Validation	Invalid inputs shall display validation messages	Medium	Business Rules (General)	Functional testing – verify validation messages appear for all invalid input scenarios

5.4 Security Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-S-001	Authentication	Users shall login using username and password	High	System Security Needs	Security testing – verify login with valid and invalid credentials
NFR-S-002	Password Security	Passwords shall be encrypted before storage	High	System Security Needs	Code review and database inspection – verify passwords are hashed in database
NFR-S-003	Role-Based Access Control	Users shall access only authorised modules	High	Business Rules (BR-002, BR-006)	Security testing – verify each role can access only permitted modules
NFR-S-004	Session Timeout	Sessions shall expire after 30 minutes of inactivity	High	System Security Needs	Functional testing – verify session expires after 30 minutes of inactivity
NFR-S-005	Secure Communication	System communication shall use HTTPS	High	Communication Standards	Security testing – verify all API communication uses HTTPS

NFR-S-006	Input Validation	User inputs shall prevent SQL Injection and XSS attacks	High	Security Standards	Penetration testing – verify system resists SQL Injection and XSS attack attempts
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5.5 Scalability Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-SC-001	Concurrent Users	System shall support at least 100 concurrent users	Medium	Business Growth Needs	Load testing – simulate 100 concurrent users and verify system performance
NFR-SC-002	Data Growth	Database shall support future growth without performance degradation	Medium	Long-term Operational Needs	Performance testing – verify query performance as database records increase

5.6 Maintainability Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-M-001	Modular Architecture	Frontend, backend, and database shall remain separated	High	Development Best Practices	Code review – verify separation of frontend (React), backend (Spring Boot), and database (MySQL)
NFR-M-002	API Documentation	APIs shall be documented properly	Medium	Development Standards	Documentation review – verify all APIs are documented in Postman or equivalent tool
NFR-M-003	Logging	System logs shall be maintained for troubleshooting	Medium	Operational Needs	System review – verify server-side logs capture errors and key system events

5.7 Portability Requirements

ID	Requirement	Acceptance Criteria	Priority	Source	Verification Method
NFR-PO-001	Browser Compatibility	System shall support Chrome, Edge, Firefox, and Safari	High	System Constraints (CON-001)	Compatibility testing – verify system functions correctly on all four browsers
NFR-PO-002	Operating System Compatibility	System shall work on Windows and Linux environments	Medium	System Constraints (CON-001)	Compatibility testing – verify system functions on Windows and Linux servers
NFR-PO-003	Database Portability	Database shall support MySQL deployment	Medium	System Constraints (CON-005)	Environment testing – verify system deploys successfully with MySQL database

Section 6 — Alternative Solutions & Feasibility Study

6.1 Alternative Solutions

This section identifies possible alternative approaches considered before selecting the Vani Tuition Centre System solution.

6.1.1 Existing Manual System

Advantages:

- No software development cost
- Simple operational process
- Minimal technical knowledge required

Limitations:

- High administrative workload
- Increased possibility of human errors
- Difficult attendance tracking
- Delayed communication with parents
- Inefficient assignment management
- Difficult report generation
- Limited scalability

Decision: Rejected due to operational inefficiencies and limited automation capability.

6.1.2 Spreadsheet-Based Management System

Advantages:

- Low implementation cost
- Easy data entry
- Basic reporting capability

Limitations:

- Limited security controls
- Difficult assignment submission handling
- No QR attendance support
- No AI learning support
- No automated SMS notification functionality
- High maintenance effort

Decision: Rejected because spreadsheet systems cannot provide centralised automation and advanced functionality.

6.1.3 Separate Independent Systems

Advantages:

- Specialised functionality
- Existing solutions available

Limitations:

- Data fragmentation
- Multiple system maintenance requirements
- User inconvenience
- Increased operational complexity
- Higher integration difficulties

Decision: Rejected because operational data should remain centralised.

6.1.4 Proposed Integrated Web-Based Solution (EduVani)

Advantages:

- Centralised management
- Reduced manual work
- Automated SMS notifications
- QR attendance automation
- AI-assisted learning support
- Improved report generation
- Better operational efficiency
- Improved parent communication
- Better scalability

Limitations:

- Initial development effort required
- Internet dependency
- System maintenance requirements

Decision: Selected as the final solution.

6.1.5 Solution Comparison Table

Feature	Manual	Spreadsheet	Independent	Proposed
Student Management	Limited	Moderate	Moderate	Yes
QR Attendance	No	No	Partial	Yes
Assignment Handling	No	Limited	Yes	Yes
SMS Notifications	No	No	Partial	Yes
Marks Management	Manual	Moderate	Yes	Yes
AI Learning Support	No	No	No	Yes
Centralised Database	No	Partial	No	Yes
Report Generation	Difficult	Moderate	Moderate	Yes

Parent Communication	Manual	Manual	Partial	Yes
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6.2 Feasibility Study

The feasibility study evaluates whether the Vani Tuition Centre System can be successfully developed and implemented considering technical, operational, economic, and scheduling aspects.

6.2.1 Technical Feasibility

Factor	Evaluation
Frontend Technology	React framework supports responsive web application development
Backend Technology	Spring Boot provides scalable backend architecture
Database Technology	MySQL supports reliable relational database management
QR Attendance Integration	Technically achievable through QR validation mechanisms
SMS Notification Service	External SMS API integration is feasible
AI Learning Support	AI chatbot integration can be implemented using AI services
Hosting Environment	Cloud or local server deployment is possible

Conclusion: The project is technically feasible because required technologies and development resources are available.

6.2.2 Economic Feasibility

Expected Costs:

Cost Category	Description
Development Cost	Software development effort
Server Cost	System hosting infrastructure
SMS Service Cost	SMS notification integration expenses
Maintenance Cost	System updates and operational maintenance
Internet Infrastructure	Network connectivity expenses

Expected Benefits:

Benefit	Description
Reduced Manual Work	Administrative workload reduction
Improved Efficiency	Faster tuition centre operations
Better Parent Communication	Automated SMS notifications
Reduced Errors	Minimised manual record mistakes
Improved Student Monitoring	Better attendance and marks tracking

Conclusion: The expected operational benefits justify implementation costs; therefore, the project is economically feasible.

6.2.3 Operational Feasibility

Factor	Evaluation
User Friendliness	Simple interface design improves usability
Staff Adaptation	Teachers and administrators can learn system usage
Parent Communication	SMS notifications improve operational communication
Attendance Management	QR attendance improves operational efficiency
Assignment Handling	Digital submission improves academic workflow

Conclusion: The system is operationally feasible because users can effectively utilise proposed functionalities.

6.2.4 Schedule Feasibility

Phase	Estimated Duration
Requirement Gathering	1 Week
System Design	2 Weeks
Frontend Development	3 Weeks
Backend Development	4 Weeks
Database Development	2 Weeks
System Integration	2 Weeks
Testing	2 Weeks
Deployment	1 Week

Conclusion: Project implementation is feasible within the planned academic development schedule.

6.2.5 Feasibility Summary

Feasibility Type	Result
Technical Feasibility	Feasible
Economic Feasibility	Feasible
Operational Feasibility	Feasible
Schedule Feasibility	Feasible

Final Assessment: The Vani Tuition Centre System is considered feasible for implementation based on technical capability, financial viability, operational suitability, and project scheduling constraints.

END OF SOFTWARE REQUIREMENT SPECIFICATION DOCUMENT